

JUAN MANUEL LORENZO NAVEIRO

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EMPLOYMENT

- Postdoctoral Research Associate and Lecturer of Mathematics, Department of Mathematics, University of Oklahoma (August 2025 – Present).
- FPI Fellow of the Spanish Ministry of Science and Innovation, Department of Mathematics, Universidade de Santiago de Compostela (September 2021 – August 2025).
- Research Technician, Department of Mathematics, Universidade de Santiago de Compostela (January – August 2021).

EDUCATION

- PhD in Mathematics, Universidade de Santiago de Compostela, July 2025.
Thesis: *Submanifolds and actions on homogeneous manifolds*.
Advisor: José Carlos Díaz Ramos.
Qualification: sobresaliente *cum laude*.
- Master's degree in Mathematics, Universidade de Santiago de Compostela (Average grade: 10/10) (September 2020 – July 2021).
- Graduate in Mathematics, Universidade de Santiago de Compostela (Average grade: 9.96/10) (September 2016 – July 2020).

RESEARCH INTERESTS

- Submanifold geometry on manifolds with symmetry: Lie groups, isometric actions on homogeneous and symmetric spaces, polar actions, homogeneous foliations, isoparametric hypersurfaces, totally geodesic and minimal submanifolds.
- Possible kinematics: kinematical Lie algebras and non-Lorentzian homogeneous spacetimes.

RESEARCH ARTICLES

- J. C. Díaz-Ramos, J. M. Lorenzo-Naveiro: Codimension two polar homogeneous foliations on symmetric spaces of noncompact type, *Adv. Math.* **472** (2025), 110295.

PREPRINTS

- J. C. Díaz-Ramos, J.M. Lorenzo-Naveiro: Polar homogeneous foliations on symmetric spaces of rank one, [arXiv:2606.27912](https://arxiv.org/abs/2606.27912) (submitted).
- J. M. Lorenzo-Naveiro, A. Rodríguez-Vázquez: Totally geodesic submanifolds of the homogeneous nearly Kähler 6-manifolds and their G_2 -cones, [arXiv:2411.11261](https://arxiv.org/abs/2411.11261) (submitted).
- J. M. Lorenzo-Naveiro, I. Solonenko: Sections of polar actions, [arXiv:2111.05280](https://arxiv.org/abs/2111.05280).

TALKS AT CONFERENCES AND WORKSHOPS

- Soul and Shape: Metric Structures and Geometric Analysis, University of Oklahoma (USA), 8 – 10 April 2026. *Nearly Kähler geometry and totally geodesic submanifolds*.
- Symmetry and Shape, Universidade de Santiago de Compostela (Spain), 3 – 7 November 2025. *Polar homogeneous foliations on hyperbolic spaces*.
- Meeting of young researchers of the Spanish Network of Geometric Analysis, Universidade de Santiago de Compostela (Spain), 28 – 29 November 2024. *Totally geodesic submanifolds of nearly Kähler and G_2 -manifolds*.
- Symmetry and Shape, Universidade de Santiago de Compostela (Spain), 23 – 27 September 2024.

Nearly Kähler geometry and totally geodesic submanifolds.

- Geometry day, Universidade de Santiago de Compostela, 11 April 2024. *Totally geodesic submanifolds in homogeneous six-dimensional nearly Kähler spaces.*
- Differential Geometry and its Applications, University of Hradec Králové, 17 – 22 July 2022. *Polar homogeneous foliations on symmetric spaces of noncompact type.*
- Workshop on manifolds with symmetries, Universität Stuttgart, 28 – 31 March 2022. *Polar actions on some noncompact symmetric spaces.*
- Symmetry and Shape, Universidade de Santiago de Compostela (Spain), 15 – 18 October 2021. *Polar homogeneous foliations on $SL(3, \mathbb{R})/SO(3)$.*

POSTERS

- Symmetry and Shape, Universidade de Santiago de Compostela (Spain), 13 – 16 October 2022, *Codimension two polar homogeneous foliations on noncompact symmetric spaces.*

TALKS AT SEMINARS

- Geometry and Topology Seminar, University of Oklahoma (USA), 6 May 2026, *Symmetric spaces and holonomy.*
- Geometry and Topology Seminar, University of Oklahoma (USA), 4 March 2026, *Polar homogeneous foliations on hyperbolic spaces.*
- Geometry, Topology and Applications Seminar, Universidade do Minho (Portugal), 12 January 2026, *Totally geodesic submanifolds of nearly Kähler and G_2 -manifolds.*
- Geometry and Topology Seminar, University of Oklahoma (USA), 12 November 2025, *Totally geodesic submanifolds of nearly Kähler and G_2 -manifolds.*
- Graduate Student Seminar, University of Oklahoma (USA), 6 October 2025, *Polar actions on symmetric spaces.*
- Seminar of the Institute of Geometry and Topology, Universität Stuttgart (Germany), 14 November 2024, *Totally geodesic submanifolds of nearly Kähler and G_2 -manifolds.*
- Pure mathematics seminar, University of Queensland (Australia), 19 July 2024, *Polar actions on symmetric spaces of noncompact type.*
- Differential Geometry seminar, University of Adelaide (Australia), 31 May 2024, *Nearly Kähler geometry and totally geodesic submanifolds.*
- Geometry seminar, KU Leuven (Belgium), 16 February 2024, *Totally geodesic submanifolds of the homogeneous nearly Kähler 6-manifolds.*
- Geometry seminar, KU Leuven (Belgium), 19 May 2023, *Polar homogeneous foliations on noncompact symmetric spaces.*
- Symmetric spaces seminar, online seminar, 2 February 2022, *Symmetric spaces of noncompact type I: The Cartan and Iwasawa decompositions.*
- Symmetric spaces seminar, online seminar, 1 December 2021, *Introduction to polar actions.*
- Introduction to research seminar, Universidade de Santiago de Compostela (Spain), 24 February 2021, *The geometry of the Heisenberg group.*

RESEARCH STAYS

- Universität Stuttgart (Germany), 11/11/2024 – 15/11/2024 (1 week).
- University of Queensland (Australia), 29/04/2024 – 26/07/2024 (3 months).
- University of Adelaide (Australia), 28/05/2024 – 31/05/2024 (1 week).
- KU Leuven (Belgium), 29/01/2024 – 16/02/2024 (3 weeks).
- University of Edinburgh (United Kingdom), 19/06/2023 – 28/07/2023 (6 weeks).
- KU Leuven (Belgium), 01/05/2023 – 26/05/2023 (1 month).
- Instituto de Ciencias Matemáticas – ICMAT-CSIC (Spain), 01/09/2020 – 03/11/2020 (2 months).

GRANTS AND AWARDS

- Grant of the National Programme for the Formation of Research Personal (FPI). Ministry of science and innovation, September 2021 – Present.
- Severo Ochoa – Introduction to Research Grant. ICMAT, CSIC (Spain), September 2020 – November 2020.
- Graduation Award. Xunta de Galicia (Spain), 2020.
- Extraordinary graduation award. Universidade de Santiago de Compostela (Spain), 2020.
- Collaboration grant. IMAT, Universidade de Santiago de Compostela (Spain), year 2019–2020.
- Academic excellence award. Xunta de Galicia, year 2018 – 2019.
- Academic excellence award. Xunta de Galicia, year 2017 – 2018.

ADVISORSHIP OF STUDENTS

- Undergraduate thesis of Marta Alonso Mourelle, titled *Representations of finite groups*. Universidade de Santiago de Compostela, 14/07/2026.
- Undergraduate thesis of Sergio Ferreiro Oroña, titled *Special relativity*. Universidade de Santiago de Compostela, 14/07/2026.
- Undergraduate thesis of Diego Asín de Alcalá, titled *Clifford algebras and spin groups*. Universidade de Santiago de Compostela, 16/07/2025.
- Undergraduate thesis of Marcos Montaña Fernández, titled *Topological groups*. Universidade de Santiago de Compostela, 17/07/2024.
- Undergraduate thesis of Tomás Rodríguez Vázquez, titled *Global theory of curves*. Universidade de Santiago de Compostela, 18/07/2023.

SERVICE

- Referee for the journal *Differential Geometry and its Applications*.
- Referee for the journal *Results in Mathematics*.
- Referee for the journal *Mediterranean Journal of Mathematics*.

OTHER PUBLICATIONS

- J. M. Lorenzo-Naveiro, *Submanifolds and actions on homogeneous manifolds* (PhD thesis), in *Publicaciones del Departamento de Geometría y Topología* **161**, Universidade de Santiago de Compostela (2025).
- V. Cora-Calvo, J. M. Lorenzo-Naveiro, A. Pérez-Rodríguez, M. Picos-Maiztegui, M. Vidal-García (Editors), *As matemáticas do veciño: Actas do Seminario de Iniciación á Investigación 2023 – 2024*, CITMAGA (2025). ISSN: 2171-6536.
- J. M. Lorenzo-Naveiro, *Polar homogeneous foliations on symmetric spaces*, in *Publicaciones del Departamento de Geometría y Topología* **149**, Universidade de Santiago de Compostela (2022) (This is my M. Sc. thesis).
- J. M. Lorenzo-Naveiro, *The geometry of the Heisenberg group*, in *As matemáticas do veciño, Actas do Seminario de Iniciación á Investigación*, Institute of Mathematics, Universidade de Santiago de Compostela (2021), ISSN: 2171-6536 (Expository article on sub-Riemannian geometry).

ATTENDANCE TO COURSES

- *Riemannian manifolds, CR manifolds, and statistical manifolds*, Unviersidade de Santiago de Compostela, October 2024.
- *Differential topology*, Universidade de Santiago de Comostela, February – May 2024.
- *The Ambrose-Singer Theorem*, Universidade de Santiago de Compostela, October – November 2021.
- *Representation theory of compact groups*, Universidade de Santiago de Compostela, September – December 2020.
- *Knots and braids*, ICMAT, September 2020.
- *Homotopy of embedding spaces*, ICMAT, September 2020.

- *Fiber bundles and connections*, Universidade de Santiago de Compostela, February – May 2020.
- *International Masterclasses – hands on particle physics*, Universidade de Santiago de Compostela, March 2016.

ORGANIZED EVENTS

- Member of the scientific committee for the conference *7th BYMAT Conference, Bringing Young Mathematicians Together*, Universidad de Sevilla, 17 – 20 November 2025.
- Co-organizer of the meeting *Peregrinando: traxectorias internacionais nas matemáticas e na física*, Universidade de Santiago de Compostela, 20 December 2024.
- Co-organizer of the conference *Matemáticas: habelas, hainas! 2023*, Universidade de Santiago de Compostela, 23 November 2023 (conference for the dissemination of Mathematics).
- Co-organizer of the PhD students seminar *Seminario de Iniciación á Investigación* and editor of the proceedings, Universidade de Santiago de Compostela, September 2023 – May 2024.

TEACHING EXPERIENCE

- *MATH 3333: Linear Algebra*. University of Oklahoma, Spring 2026.
- *MATH 1823: Calculus and Analytic Geometry I*. University of Oklahoma, Fall 2025.
- *Topology of surfaces*. Degree in Mathematics (3rd year), Universidade de Santiago de Compostela, 2023/2024.
- *General topology*. Degree in Mathematics (3rd year), Universidade de Santiago de Compostela, 2022/2023, 2023/2024.
- *Global theory of surfaces*. Degree in Mathematics (3rd year), Universidade de Santiago de Compostela, 2021/2022.
- *Mathematics and Statistics II*. Degree in Pharmacy (1st year), Universidade de Santiago de Compostela, 2021/2022, 2022/2023, 2023/2024.
- *Mathematics*. Degree in Biotechnology (1st year), Universidade de Santiago de Compostela, 2021/2022, 2022/2023, 2023/2024.

LANGUAGES

- Native languages: Spanish and Galician.
- Fluent in English (Cambridge CAE, Level C2).
- Basic skills in French (DELF A2).